

Ninad Deshmukh

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Education

University of Maryland

Master of Science in Robotics

College Park, MD

Jan 2026 – Dec 2027 (Expected)

Coursework: Robot Modeling, Control of Robotic Systems, Human-Robot Interaction

MIT ADT University

B.E. Electronics and Computer Engineering

Pune, India

Aug 2019 – Jul 2023

Coursework: Electronics Circuit Analysis, Embedded Systems, IoT, Machine Learning, Computer Networks

Technical Skills

Languages: Python, C/C++, MATLAB

Robotics: UAV Autonomy, Robot Perception, Motion Planning, Sensor Fusion, Navigation, Obstacle Avoidance, Localization, Mission Planning, System Integration

Embedded Systems: Arduino, ESP32, Raspberry Pi, Pixhawk (ArduPilot), I2C, UART, CAN, PCB Design, Circuit Design, Hardware Debugging

Computer Vision / AI: OpenCV, TensorFlow, Image Processing, Object Detection, Deep Learning, Neural Networks, Vision-Based Navigation

Drone Technologies: DJI Enterprise Ecosystem, Flight Testing, Telemetry, Precision Landing, Geofencing, BVLOS, Payload Integration, Dock Integration

Software / Tools: Linux, Git, Docker, REST APIs, FastAPI, EasyEDA, Autodesk Fusion, Mission Planner

Engineering: System Validation, Benchmarking, Hardware-Software Integration, Root Cause Analysis, Technical Documentation, Data Visualization

Professional Experience

FlytBase

Aug 2023 – Jan 2025

UAV Systems Engineer

Pune, India

- Planned and executed flight tests for software releases and hardware integrations across DJI Dock 1/2, Matrice 30, and M350 platforms, validating autonomous navigation, obstacle avoidance, real-time telemetry, precision landing, failsafe recovery, payload control, and geofence compliance
- Reduced system downtime by **15%** by leading troubleshooting efforts for malfunctioning UAV systems and investigating reported in-field incidents
- Performed mechanical and electrical integration of parachute recovery systems, payloads, and sensor modules, validating designs through bench testing and field trials
- Defined and validated customer workflows for fleet management, payload control, and BVLOS (Beyond Visual Line of Sight) operations, including third-party docking integrations (Hextronics, Heisha, IDI-Ployer)
- Improved ground control software UI/UX through field testing and real-world troubleshooting; authored SOPs for remote mission execution
- Collaborated with Product Management, Customer Success, and Engineering teams to convert customer feedback into shipped improvements, including bug fixes and third-party payload integrations

FlytBase

Feb 2023 – Jul 2023

UAV Systems Engineer Intern

Pune, India

- Assisted in flight planning, hazard analysis, and software validation testing for autonomous UAV operations
- Collaborated with Product Management, Customer Success, and Web teams to incorporate customer feedback into product development
- Researched industry trends and competitor products to recommend feature and workflow improvements

Projects

Design and Development of Sub-250g Autonomous Drone

Dec 2023 – Present

ArduPilot, Pixhawk, JeVois Camera, Python

- Designed and developed a sub-250g autonomous UAV using onboard vision processing and autonomous flight control.
- Integrated Pixhawk, JeVois camera, IR communication modules, and custom electronics while performing system validation.
- Optimized ArduPilot parameters and designed a real-time communication architecture for collaborative reconnaissance missions.

Autonomous Unmanned Aerial Vehicle

Jun 2022 – May 2023

Pixhawk, Embedded Systems, Wireless Communication

- Built an autonomous UAV supporting payloads for mapping, inspection, wildlife monitoring, and environmental data collection.
- Performed hardware integration, assembly, Pixhawk configuration, and flight-controller tuning for reliable operation.
- Implemented telemetry and payload data transmission for remote monitoring under varying environmental conditions.

AI-Powered Smart Mirror

Jan 2022 – May 2022

Python, IoT, Embedded Electronics, AI

- Developed an AI-enabled smart mirror integrating voice recognition, IoT connectivity, weather, reminders, and news services.
- Designed hardware architecture including circuit design, power management, and embedded system integration.
- Combined sensing, connectivity, and AI-driven functionality into a unified consumer-focused intelligent system.

Certifications

Robotics Operating System-2 (ROS2)

Android Development